Generative AI in Story Co-creation to Enrich the Polariscope Platform

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Abstract

This research focuses on the integration of Generative Artificial Intelligence (GenAI) into the Polariscope platform, with the objective of enriching the creation and co-creation of stories and contents. The aim is to develop and analyze an intelligent assistant capable of supporting users in writing, generating descriptions for content, and automatically recommending media, utilizing the platform's existing archives. The study seeks to understand the extent to which AI can contribute to the creativity, efficiency, and quality of the generated stories, as well as to assess user interest and engagement with this type of creative assistance. Following a Design-Based Research (DBR) methodology, an initial prototype was developed and evaluated through two consecutive cycles of usability testing. Preliminary results indicate that iterative UI/UX refinements, such as adjustable creativity modes, proactive media recommendations and a dedicated mobile interface, significantly enhanced system intuitiveness and user agency. Ultimately, these early findings suggest that the tailored AI assistant has the potential to help users overcome creative blocks and speed up the narrative construction process, mitigating user frustration and impatience, thereby paving the way for the upcoming final tests.

CCS Concepts

• **Human-centered computing → Collaborative and social computing systems and tools**; **Interactive systems and tools**.

Keywords

Artificial Intelligence, Co-creation, Digital Storytelling, Creativity, Intelligent Assistant.

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1 Introduction

The preservation of cultural heritage and shared events is essential. In the digital age, this relies heavily on the media records citizens capture while experiencing the events. These media records can be transformed into narratives by means of publication on social media or specialized storytelling platforms. Digital storytelling platforms empowers citizens to share and co-create by weaving isolated personal memories into rich, collective histories, ensuring they remain accessible and engaging.

Rooted in this collaborative vision, Polariscope is a research and development project focused on the design and implementation of a digital platform dedicated to collaborative storytelling. It serves as an interactive ecosystem where citizens and cultural heritage institutions converge to share archives and memories. The primary purpose of this solution is to promote the preservation of local and global history through content upload and co-creation of narratives, allowing collective memory to be shared, enriched, and reused in an open access digital environment. It is expected to contribute to increasing user engagement with cultural and historical events by enabling audiences to extend their experience beyond the events themselves through immersion in related media enriched collaborative narratives.

To understand the interaction dynamics within this platform, it is fundamental to analyze its data architecture, which is structured into three interdependent hierarchical levels: Projects, Stories, and Contents. At the top of this hierarchy is the Project, acting as a central repository or thematic page dedicated to a specific event, such as a historical milestone, a personal occurrence, or a group celebration. This level serves as a container for all information associated with a theme, allowing the creator to define privacy levels and assign collaborative roles to other members (such as editors or administrators), thereby reinforcing the platforms' collective character. Descending the structure, we find the Stories, which represent the platforms' narratives. This is the space where memories gain context through a dynamic page that articulates text, media, and challenges (call to action for content uploads). At this level, users become narrators, documenting their personal or collective experiences. Finally, the fundamental elements that make the platform feel alive, are the Contents: photographs, videos, audios, or text citations (other media formats, including 360 videos, may be included in future versions of the platform). Each uploaded content is enriched with metadata that facilitates its organization and reuse, allowing it to gain life in multiple areas of the application, from project galleries to integration into stories created by other users.

Despite relying on traditional media formats, the Polariscope platform aims, through its collaborative storytelling approach, to contribute to the immersion of participants in events or celebrations. Users may engage by sharing content or collaborating in content curation activities. Additionally, the platform provides multiple consumption formats, a feed mode and a presentation mode for deeper immersion into stories.

However, in contrast to the robustness of this structure, observations from previous testing phases of the Polariscope platform revealed a critical problem: the cognitive and creative effort required to create stories. It was identified that users prefer to adopt a passive role, limiting themselves to consuming information or uploading isolated content and avoiding the construction of complete stories due to the time, patience, and creativity they demand. This barrier reduces of the platform's narrative richness.

Faced with this need, there was an opportunity to integrate Generative Artificial Intelligence (GenAI) as a creative assistant. This necessity had been the subject of a previous study regarding automatic generation of stories from existing content within Polariscope projects [2]; however, during that phase, the platform was still under development, forcing tests to be conducted outside the real ecosystem. Furthermore, the limitations of available AI technologies at the time (such as the GPT-3.5 model) restricted the scope of the solutions. More importantly, the ultimate goal was not merely to propose fully automated stories, but rather to encourage users to actively engage in the authoring process, ensuring a human-driven AI-assisted, co-creation experience.

Nevertheless, that preliminary work was fundamental in identifying a critical conclusion: the rigidity of AI parameters does not work for all users. The preference for the level of creativity (controlled by the "Temperature" parameter) varies significantly across story contexts. Based on this insight, the current research proposes the implementation of three distinct AI modes to allow users to select the appropriate tone for their specific narrative needs: a Creative Mode (high temperature) that generates emotional and expressive narratives for festive events; an Objective Mode (low temperature) that produces formal, fact-based texts for rigorous historical accounts; and an Automatic Mode (balanced temperature) serving as the default setting to mediate between creative flair and factual accuracy.

2 Purposes and Objectives

The present research investigates means to facilitate the story creation process, mitigating the effort required to construct complex narratives and articulate multimedia contents, thereby aiming to solve the problem regarding the users' lack of time or creativity. The study proposes to explore the potential of Generative Artificial Intelligence (GenAI) as a creative assistant, capable of supporting users in writing, formulating descriptions, and automatically creating stories, promoting a more accessible and higher-quality co-creation experience.

The investigation aims to evaluate how integrating AI can improve narrative quality and user participation on the platform, without compromising the authenticity and authorship of the content. In this sense, and to understand if the intervention of this assistant optimizes the creation process, the following research question was formulated:

"Can a generative AI-supported assistant significantly contribute to the story creation process and its perceived quality?"

The development of the research is guided by the general objective of developing a tool to be integrated into the Polariscope platform to facilitate and improve the story creation process through the use of AI. To detail this main goal, four specific objectives were established: to specify functional and interaction requirements for the implementation of a story creation assistant; to develop an AI assistant capable of generating stories from existing content that guarantees creative control by the users; to evaluate story creators' perception regarding the quality and satisfaction of the support provided by the assistant on the platform; and to identify the control expectations users have when utilizing the assistant.

3 State of the Art

Digital Storytelling (DS) combines multimedia tools to create engaging and interactive narratives, bridging the tradition of oral narration with modern digital expressive possibilities [7, 11]. In creative domains, collaboration acts as a catalyst for innovation enabling the continuous sharing of knowledge and skills within co-creation platforms [8]. However, designing platforms that effectively support collaborative creativity faces significant challenges, as their efficacy depends on the ability to structure participation and facilitate the exchange of ideas [1, 4]. Studies show that the success of these platforms is frequently linked to the presence of facilitators or support mechanisms that guide the creative process and minimize group work difficulties [10].

Generative AI (GenAI) distinguishes itself from traditional AI by its ability to synthesize new, original content rather than merely interpreting or classifying existing data [3, 9]. The evolution of Large Language Models (LLMs) has transformed AI from a simple algorithm into an active assistant or creative partner in the content creation process [6]. In Human-AI co-creation, the AI acts as a "scaffolding" tool, suggesting ideas and guiding the user through creative stages, thereby increasing productivity and originality [5].

A benchmarking analysis of current market tools revealed a significant gap: while tools like Sudowrite excel in producing fiction, and Canva Magic Write focuses on multimodal creation for social media, there is a lack of collaborative platforms that integrate AI specifically to support the contextual remixing and enrichment of real, user-generated archival content. It is important to clarify that this benchmarking analysis focused solely on text generation capabilities, evaluating how these models handle narrative construction and factual coherence. This gap confirms the relevance of developing a tailored solution (PolarisAI) for the Polariscope platform.

4 PolarisAI: Co-Creation Assistant

To address the technological gap identified in the market analysis and to cater specifically to Polariscopes' unique collaborative environment, PolarisAI was developed. PolarisAI is a tailored, context-aware Generative AI assistant seamlessly integrated into the platform's story creation interface.

The concept for this tool emerged directly from observing user behavior during the early testing phases of Polariscope. As previously noted, the cognitive and creative demands of articulating fragmented multimedia archives into cohesive narratives frequently overwhelmed users, leading to "writer's block" or passive consumption (lurking). The need for an intelligent solution had been the subject of a preliminary study exploring the automatic generation of stories [2]. However, that early iteration was constrained by the platform's developmental stage and the limitations of the available AI technologies at the time, such as the GPT-3.5 model.

Consequently, PolarisAI was designed to overcome these historical and technological barriers. It was created not to generate fully automated stories that bypass the user, but rather to act as a supportive "scaffolding" partner. The ultimate reason for its existence is to empower users, encouraging them to actively engage in the authoring process and ensuring a human-driven, AI-assisted co-creation experience. Before detailing the evaluation of the system, it is relevant to understand the specific mechanisms of this artifact.

4.1 Objectives of PolarisAI in Polariscope

Building upon its conceptual origins, the primary objective of PolarisAI is to mitigate the cognitive load and time constraints associated with collaborative digital storytelling. It aims to support users in overcoming writer's block and structuring their narratives from isolated multimedia records, while ensuring that the human creator retains ultimate authorship and creative control over the final output.

4.2 Main Functionalities

To achieve these goals, the assistant is equipped with three core capabilities. First, it offers contextual generation, creating engaging titles, short descriptions, and narrative paragraphs that align with the projects' overarching theme. Second, it provides smart rewriting functions, that improve texts previously written by the user. Finally, it features content recommendation, which analyzes the current state of the story creation process and scans the project's archives to recommend the most relevant multimedia assets, such as photos, videos, or citations, to be added next.

4.3 Mode of Operation and Technical Implementation

PolarisAI operates through two seamless interaction paradigms designed to minimize user friction. The first paradigm consists of in-context quick buttons placed directly next to input fields, such as the title and description areas. Depending on whether the field is empty or filled, these buttons dynamically switch between "Generate" and "Rewrite," allowing users to request AI assistance with a single click without manually typing a prompt. The second paradigm is a conversational chat assistant, serving as a dedicated interface where users can provide specific instructions, such as asking the AI to write a paragraph semantically linking two distinct photos. On desktop, this chat is presented in a fixed sidebar, whereas on mobile, it is accessible via a Floating Action Button (FAB) that triggers a slide-up modal.

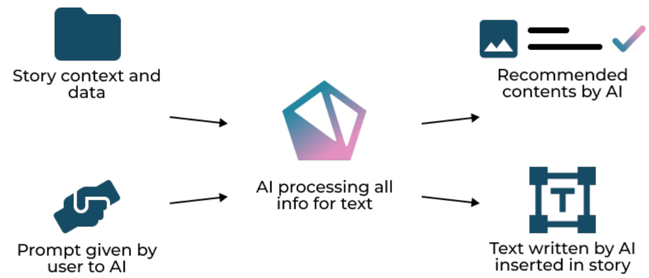

Figure 1: PolarisAI Scheme

From a technical standpoint, PolarisAI is powered by the Google Gemini API, specifically using the gemini-3-flash-preview model. To ensure the AI acts as a contextual co-author rather than a generic chatbot, strict backend prompting is implemented. Whenever a user interacts with PolarisAI, the backend dynamically compiles the user's current progress, including the project's theme, the story's title, keywords, and all previously written text blocks into a structured context prompt.

To ensure the AI's output can be easily integrated into the platform's UI, strict system instructions require the model to return responses using predefined structural tags (e.g., [[TITLE]], [[DESCRIPTION]], and [[TEXT]]) accompanied by block IDs. A custom parser in the frontend processes these tags and presents the suggestions cleanly to the user as cards. By clicking the "Accept" button, the generated text is instantly injected into the correct input field of the story. Furthermore, to manage API usage and encourage thoughtful interaction, a token system is implemented, limiting users to a maximum of 20 AI interactions per story or 12-hour cycle.

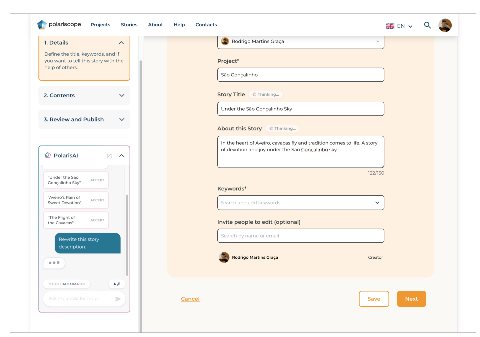

Figure 2: Initial Version of PolarisAI

Having established the conceptual design and technical implementation of PolarisAI, it became necessary to evaluate and optimize the tool within a real-world context. To achieve this, an evaluation methodology was designed, comprising successive phases of testing and refinement, which are described below.

5 Methodology

The study is structured into four main phases:

Phase 1 - Conceptualization: Definition of the concept, state-of-the-art analysis, and functional requirements specification.

Phase 2 - First Prototype: Technical development and integration of the AI assistant into the Polariscope architecture.

Phase 3 - Iteration (UX and Usability Testing): Short, individual testing sessions with users to assess usability issues, workflow flaws, and acute design needs.

Phase 4 - Final Evaluation: Large-scale quantitative evaluation (questionnaires) and in-depth qualitative interviews.

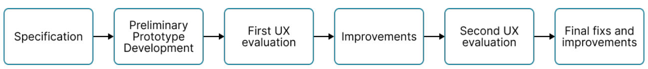

Figure 3: The PolarisAI DBR approach

Since Phase 1 and Phase 2 culminated in the development of the initial PolarisAI prototype described in the previous section, the paper reports the current progress of this research, which focuses primarily on the execution of Phase 3, aiming to refine the artefact before its final deployment and evaluation.

6 UX Iterations and Usability Testing (Phase 3)

To evaluate the integration of PolarisAI and optimize its interface, Phase 3 comprised two usability testing iterative cycles. These individual, in-person sessions were conducted in a controlled and consistent testing environment provided by the researcher. During the tests, the researcher acted as an observer and facilitator, taking notes and audio-recording the sessions for later analysis. The researcher intervened only to provide predefined hints when participants explicitly requested help or encountered a critical usability barrier. On average, each testing session lasted approximately 30 minutes.

Instead of an exhaustive and demanding task list, the evaluation followed a structured script of tasks designed to map the entire co-creation workflow. Participants were asked to locate the assistant, generate a story title, and interact with the chat interface. Furthermore, they were guided to use contextual text editing (rewriting), identify token usage limits, adjust the AI's creativity modes, and explore the content recommendation feature to seamlessly add multimedia assets to their stories.

Throughout the process, a structured observation grid was used to log performance metrics (task success) and qualitative behavioral reactions. The observer documented critical incidents and emotional responses, capturing moments of positive surprise, perceived user control, cognitive confusion (e.g., staring at the screen without knowing where to click), or frustration.

6.1 First UX Testing Cycle

The first round of UX testing was conducted with three users, all of whom were software developers and AI specialists. The decision to inquire a first group of participants with strong understanding of how AI works was made considering the goal of identifying technical errors and identifying awkward user flows early in the process.

The feedback from this technical group was insightful. Participants were generally surprised and impressed by Generative AI's ability to truly understand the users' story context and provide content suggestions to continue. They also appreciated the content recommendations, made when browsing for media assets to include in a story, which successfully fetched relevant media from within the project gallery.

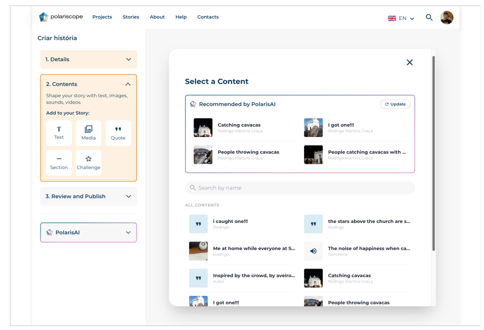

Figure 4: PolarisAI recommending contents

However, several usability issues were identified, suggesting immediate structural refinements. Regarding interface feedback and discoverability, minor UI inconsistencies caused confusion, such as a lack of clarity in interactive elements: the AI mode selector appeared as a static label rather than a dropdown, and loading states during text generation were unclear. Aditionally, the content recommendation feature was too hidden within the menus, requiring users to rely on a process of elimination to find it. Concerning layout and readability, the placement of the "Accept" button for AI-generated text reduced horizontal space, making reading difficult. Furthermore, it became clear that the fixed sidebar approach for the chat assistant was completely unfit for mobile devices, requiring a new interaction paradigm.

6.2 First Iteration and UI/UX Overhaul

Based on this feedback, significant technical and visual adjustments were made. The "pop-out" icon was redesigned to match the active UI colors. The state management was refined so that only the specific button clicked displays the "thinking..." animation. The AI Mode selector was transformed into a clear, upward-opening dropdown. The "Accept" button was relocated below the generated text, aligned to the right, significantly improving readability.

A major architectural shift occurred in the Story Editing interface. Initially, adding elements (text, media, citations) was done via a fixed bottom bar on mobile and the sidebar on desktop. To create a more intuitive flow, this was replaced with a contextual "+" button that opens and closes, situated at the bottom of the "blank page". Crucially, space at the bottom on mobile enabled the implementation of a Floating Action Button (FAB) in the bottom-right corner. Clicking this FAB opens the PolarisAI chat in a slide-up modal, solving the mobile accessibility issue. Finally, backend system instructions were adjusted to ensure the "Automatic" mode was less similar to the "Creative" mode, grounding the narratives more effectively.

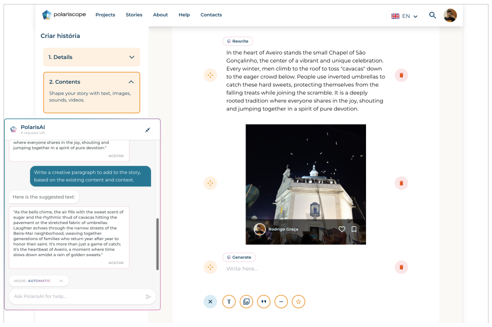

Figure 5: Interface after first iteration improvements

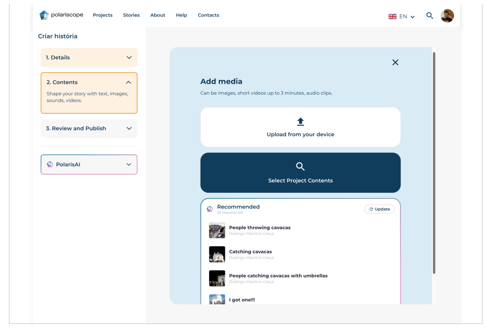

Figure 6: Interface after second iteration improvements

6.3 Second UX Testing Cycle

A second round of testing was conducted with four new participants. Unlike the first group, these participants were non-technical users from diverse backgrounds. This shift in demographics was intentional: it served to verify if the newly adjusted flows were genuinely intuitive for an average user without a technological background, and to ensure they could comfortably interact with the AI feature. To maintain consistency and enable a reliable comparison of the interface iterations, the evaluation setup, the researcher's role, and the structured task script remained the same as the one used in the first testing cycle.

The results were very positive regarding the UI changes. These users found the mobile floating button highly intuitive and praised the new placement of the "Accept" button. The AI modes were now clearly understood. The primary remaining constraint was the Content Recommendation feature. Users still struggled to find it naturally. They suggested either creating a dedicated, highly visible button for it or placing it directly in the path of the user's workflow. Another user noted that while the token limit display was aesthetically pleasing, it could easily go unnoticed.

6.4 Second Iteration and Final Polish

To solve the discoverability of recommendations, the UI was restructured. Instead of placing the AI recommendations behind a secondary modal, it was moved to the forefront. Now, when a user chooses to add media, the first modal displays four PolarisAI-recommended content cards immediately, along with options to upload local files or search the project manually.

Along with this change, the token system was also unified. Tokens are now shared between chat requests and content recommendations. Consequently, the token limit was increased from 10 to 20 per story or a 12-hour cycle. To address the tokens' limit visibility, a programmed onboarding message was added: the first time a user interacts with PolarisAI, it not only answers the prompt but also sends a "Tip" message explaining that a token was consumed and directing their attention to the counter.

Further UI polishes included adding text labels to the icons in the "+" add element menu for clarity and adding a scroll-aware animation to the mobile FAB (scrolling down collapses the button to just the logo; scrolling up expands it to show the "PolarisAI" text). Additionally, other suggested feature was added: users can now select multiple contents from the project at once to add to their story. Finally, a confirmation checkbox was added at the final publishing step, prompting the user to confirm whether PolarisAI was used, allowing the platform to transparently flag co-created stories.

7 Expansion: PolarisAI in Content Creation

Following the positive feedback regarding the storytelling assistant, a secondary issue was identified: to provide optimal content recommendations during story creation, PolarisAI relies heavily on the metadata of the project's contents. However, users frequently skip the optional "Description" field when uploading individual photos, videos, or audios, leaving the contents with incomplete contextual data.

To solve this, PolarisAI was expanded into the Content Creation section. Unlike the story interface, this integration is minimalist and features no chat interface. When a user uploads a media file (or inputs a text citation) and provides a title and keywords, a simple "Generate" button appears next to the description box.

Technically, when clicked, this triggers a dedicated endpoint that handles multimodal inputs. The Node.js backend extracts the media file, converting lightweight images or audio directly to base64, or using the Google Gemini File API for the asynchronous processing of heavier video files. To process these files, the backend uses a temporary secure local environment. Crucially, to maintain strict data privacy and storage efficiency, all uploaded media is programmatically deleted from both the local server disk and Google's cloud infrastructure the exact moment the text inference is complete.

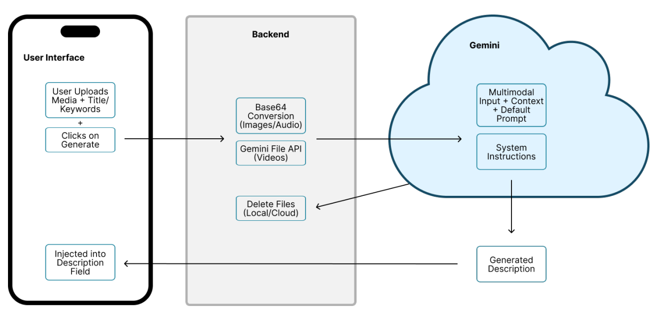

Figure 7: Multimodal processing workflow for automatic content description generation.

For the generation process, the *gemini-3-flash-preview* model consumes the media alongside a structured prompt containing the user-provided title and keywords. The system instructions strictly constrain the model to act as a factual cataloger, generating a concise, objective description under 300 characters and devoid of markdown formatting. The AI "views" the image, "watches" the video, or "listens" to the audio, and the resulting text is instantly injected into the description text area. This can then be freely edited by the user before submitting the content. This workflow drastically reduces user friction, enriching the platform's metadata organically and, consequently, improving PolarisAI's future contextual recommendations during the story creation phase.

8 Future Work (Phase 4)

While the preceding testing cycles were fundamentally focused on usability optimization and refining the user interaction flows, the prototype is now fully developed and ready for a broader assessment. The research will now move into Phase 4: Final Evaluation, where the core objective shifts to definitively assessing the assistant's relevance, the quality of its recommendations, and the overall user experience.

Future work consists in a field trial with a group of approximately 80 bachelor degree students that will use the application to create stories and access the PolarisAI assistant. A quantitative questionnaire will be distributed to the participants to access the relevance and perceived quality of the AI assistant along with Usability and User Experience evaluations relying in instruments like the System Usability Scale (SUS) and AttrakDiff. Following this, a subset of participants will be selected for in-depth qualitative interviews. These interviews will explore deeper thematic concepts, such as the user agency (whether the user feels they own the story or if the AI took over), the factual rigour of the AI when interpreting personal memories, and the overall impact of GenAI on democratizing digital storytelling versus the risk of narrative homogenization. The triangulation of these quantitative data with qualitative insights will provide a comprehensive conclusion to the research questions posed by this study.

9 Conclusion

The PolarisAI module is being developed to assist Polariscope users in the story creation process. The researchers believe that this tailored set of features will encourage an active participation within the platform by easing the creative process. With the support of an AI assistant, users will be able to build their narratives more efficiently.

The iterative testing cycles allowed for significant improvements to PolarisAI, namely regarding usability issues. The upcoming final evaluation stage will allow us to confirm whether PolarisAI is effective in assisting users in a real-world scenario. Additionally, it will be crucial to gather the users' perspectives on the intervention of such an assistant, particularly concerning its impact on their sense of control and feeling of ownership over the co-created stories.

Acknowledgments

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